

# Apache XML Parsers: Xerces2 and Xerces C++

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# Introduction

- Speaker
  - Works for IBM
  - IBM W3C DOM representative
- History of Parsers
  - First developed in IBM's Tokyo research lab
  - Maintained and expanded in California
  - Donated to Apache
  - Work continues in Toronto


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# Agenda

- Xerces2:
  - Brief overview of XNI Framework
  - Latest changes to XNI
  - Grammar Caching
  - Standards support
- Xerces C++
  - Architecture, features..


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# Xerces Native Interface (XNI)


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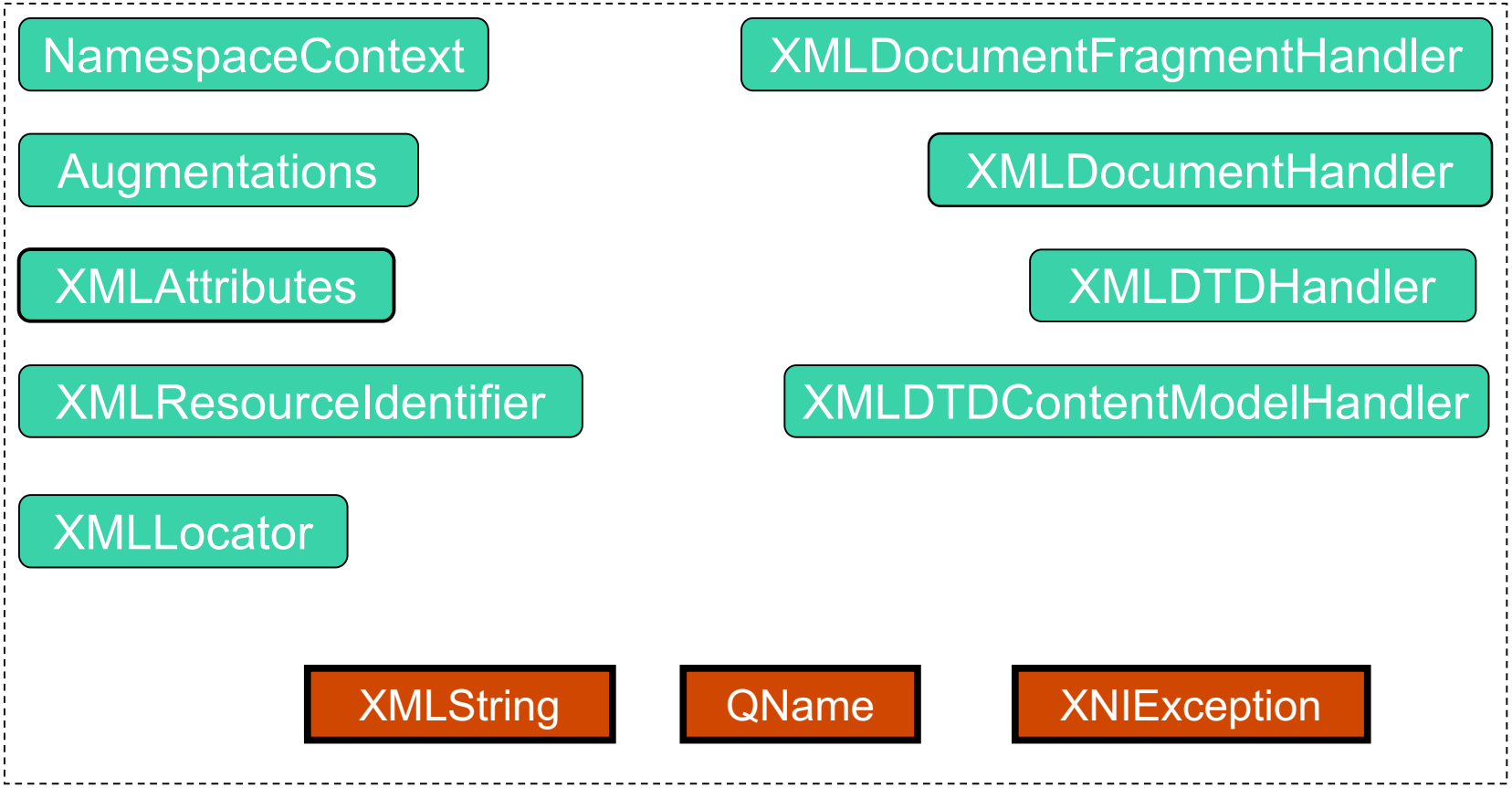


# Xerces Native Interface (XNI)

- "Streaming" Information Set
  - Similar to SAX
  - Information: XML Information Set and more
- Parser components and configurations


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# XNI Core (org.apache.xerces.xni)





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# XNI Parser Interfaces

XMLComponent

XMLComponentManager

XMLParserConfiguration

XMLPullParserConfiguration

XMLEntityResolver

XMLErrorHandler

XMLInputSource

XMLParseException

XMLConfigurationException

XMLDocumentScanner

XMLDocumentSource

XMLDocumentFilter

XMLDTDScanner

XMLDTDSource

XMLDTDFilter

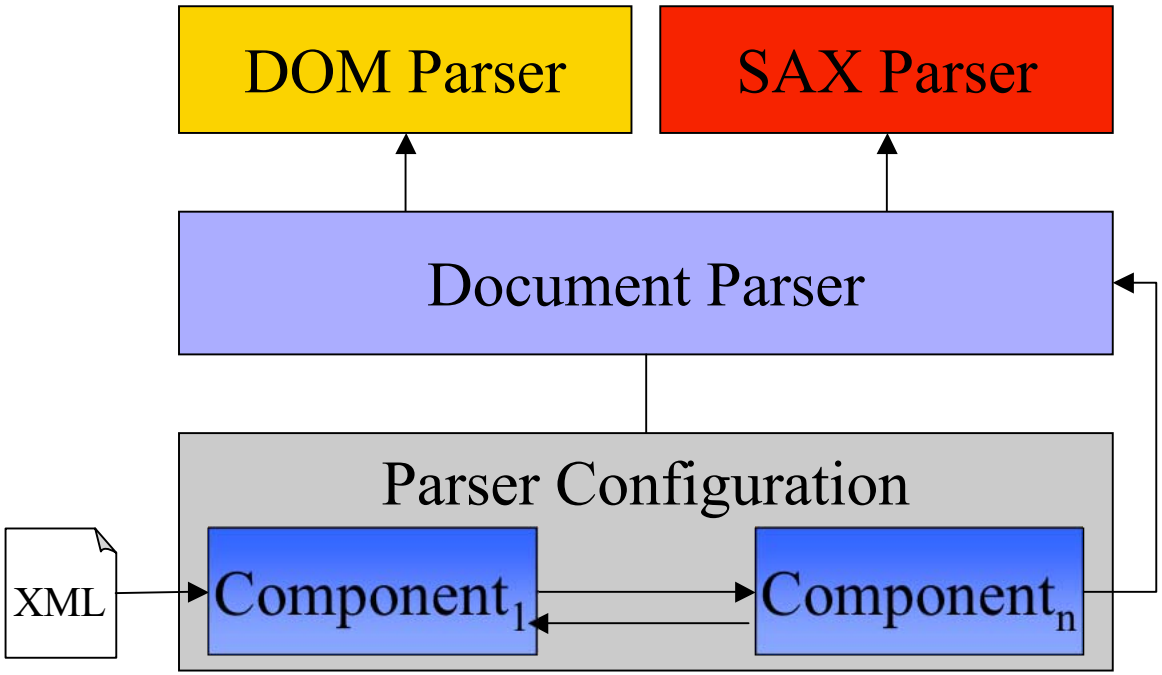
XMLDTDContentModelSource

XMLDTDContentModelFilter



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# XNI Components, Configurations, and Pipeline




* Parser pipeline and settings are specified in Parser Configuration object.

* Allows re-use of framework and components without rewriting existing code.

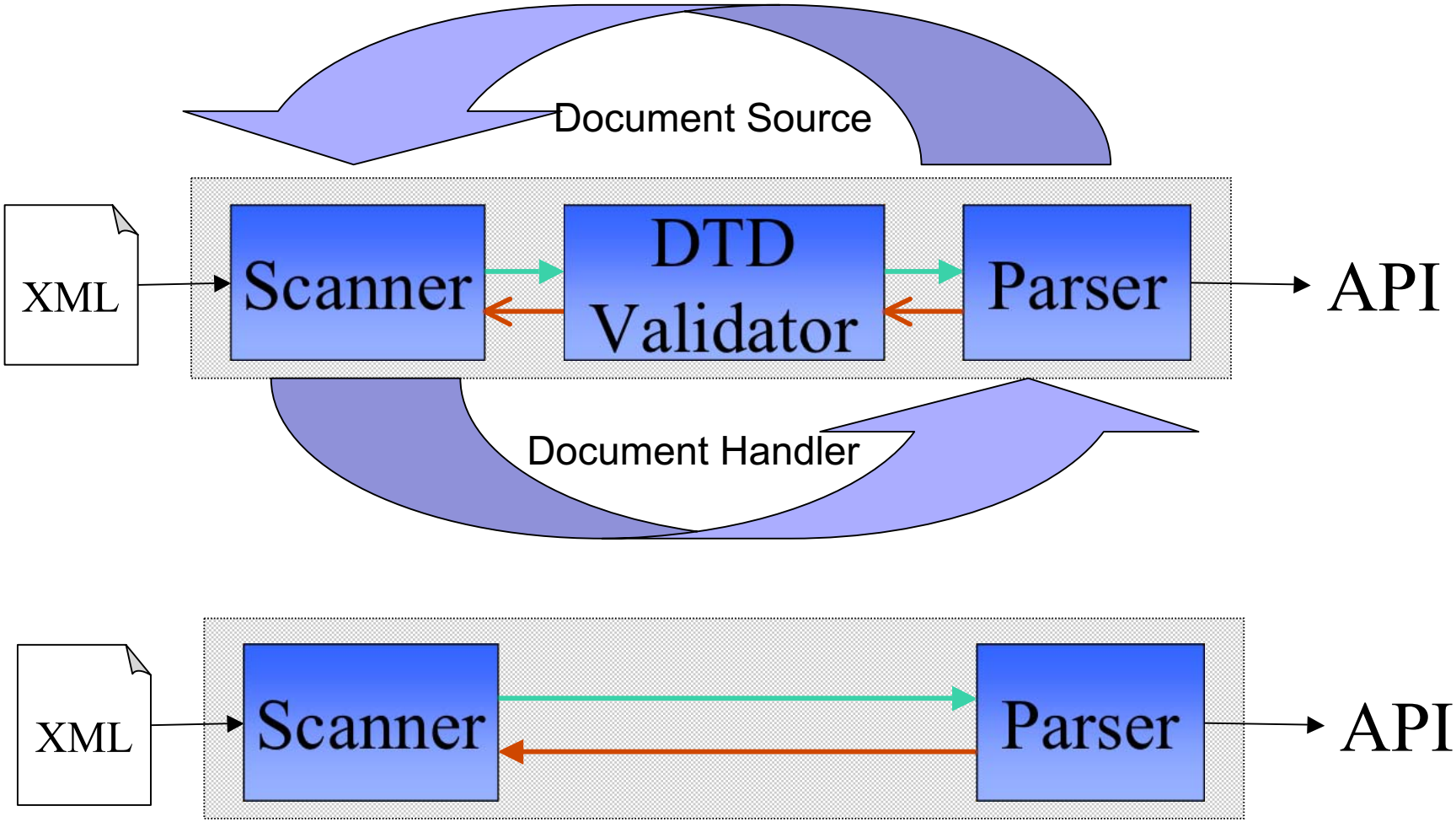

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# Latest Changes To The XNI Framework


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# Dynamic Pipeline





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# Additional XNI Changes

- Namespace and XNI pipeline
  - Eliminating startPrefixMapping and endPrefixMapping events
- XMLComponent Interface Changes
  - Adding methods for querying default values


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# Grammar API in Xerces2

- When and why to use?
  - If an application has a set of schemas used for document validation
  - When performance is a consideration


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# XNI Grammar Package

- Grammar: represents Xerces grammar object
- XMLGrammarDescription: provides general information about grammar
- XMLGrammarPool: provided by an application to the parser
- XMLGrammarLoader: API to load grammars


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# Interaction between application and Xerces

- Register the grammar pool with the parser
  - Property: http://apache.org/xml/properties/internal/grammar-pool
  - Configuration: XMLGrammarCachingConfiguration
- The validator interacts with the registered grammar pool:
  - Before the parse starts: to get initial set of grammars
  - During the parse: to get missing grammar
  - After the parse: to give to the application all the cached grammars


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# How do I start?

- Use Xerces default Grammar Pool implementation:
  - org.apache.xerces.util.XMLGrammarPoolImpl
  - Passive approach: let parser do the caching
  - Active approach: pre-parse grammars
  - Sample: xni.XMLGrammarBuilder
- Write application specific grammar pool implementation


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# Examining Grammars

- Examine XML Schemas using PSVI API
  - ***Note:*** no such API exists for DTDs!
- Access and query the post-schema-validation infoset (PSVI) **
- To access the API:
  - Cast Grammar object to XSGrammar
  - Get XSModel using toXSModel() method

** http://www.w3.org/TR/xmlschema-1/#PSVI_contributions


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# Brief Overview of the PSVI API



- **XSModel** – “A Schema”
  - Query methods for declarations
  - Access to namespace item
- **XSElementDeclaration** – element declaration
  - Value constraint, type information…
- **XSTypeInformation** – type information item
  - Base type, derivedFrom methods


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# Retrieving PSVI for an XML Document

- ElementPSVI and AttributePSVI
  - Validation information, error codes, normalized values…
- Accessing PSVI in XNI
- Accessing PSVI in SAX
  - Parser implements PSVIProvider
  - Application can pull PSVI information from the PSVIProvider


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# Accessing PSVI from DOM

- Cast Element node to ElementPSVI
- Cast Attribute node to AttributePSVI
- Use DOM Level 3 Bootstrapping mechanism

```
System.setProperty(DOMImplementationRegistry.PROPERTY,
"org.apache.xerces.dom.DOMImplementationSourceImpl");

DOMImplementationRegistry registry=
                                DOMImplementationRegistry.newInstance();

DOMImplementation i = registry.getDOMImplementation(“PSVI");
```


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# Standards and Xerces2


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# New Specifications Support

- JAXP 1.2
  - Specify schema location and schema type
- XML 1.1
  - Unicode 3.2 and beyond support
- SAX 2.0.1
- DOM Level 3


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# DOM Level 3

- Xerces2 is a reference implementation for DOM Level 3 Core and Load Save
- This API is still considered to be experimental
  - Use at your own risk
  - Build Xerces locally: “jars-dom3” ant target


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# DOM Level 3 Features

- Renaming element and attribute nodes
- Handling text content
- Moving Nodes from one document to another
- Normalizing Document
  - Normalizes Text, makes document namespace well-formed
  - Configured using DOMConfiguration parameters: “comments”, “validate”, “error-handler”


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# Validating DOM

```
// retrieve configuration
DOMConfiguration config = document.getConfig();
document.setDocumentURI(http://www.exampl.com/);

// configure normalizeDocument() method
config.setParameter("schema-type",
                    "http://www.w3.org/2001/XMLSchema");
config.setParameter("schema-location", "personal.xsd");

config.setParameter("validate", Boolean.TRUE);

// revalidate your document in memory
document.normalizeDocument();
```

* Register DOMEntityResolver

* Or specify on the documentElement xsi:schemaLocation or xsi:noSchemaLocation


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# Parser Performance

- Reuse parser instance
- Use specialized parser configurations
- Use grammar caching
- Validation
- Deferred DOM


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# Xerces-C


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# Xerces-C Agenda

- Xerces-C introduction
  - Architecture overview
  - Parser Pipeline
  - Reader Management
  - Pluggable Memory Management
- Features highlight

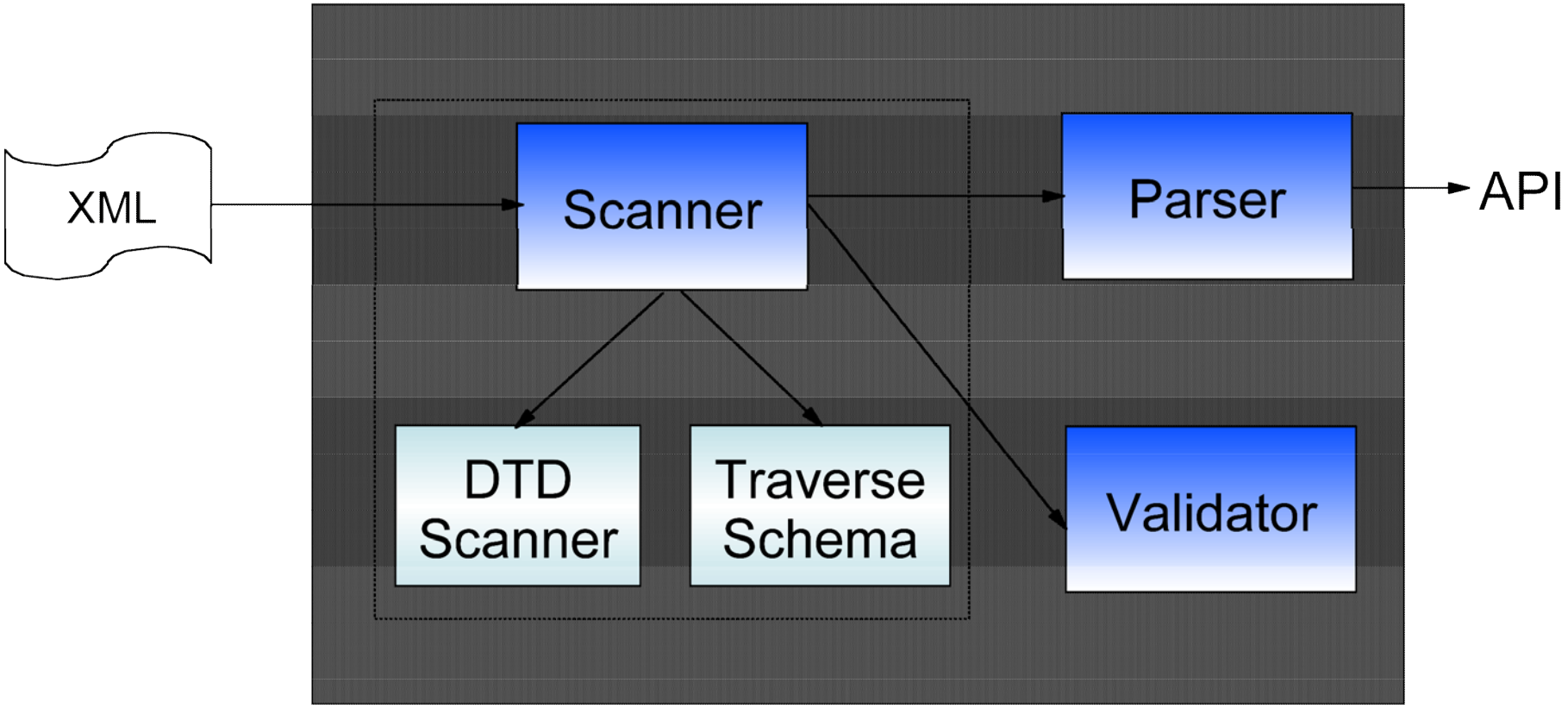

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# Xerces-C

- A validating XML parser written in a portable subset of C++
- Gives applications the ability to read and write XML data
- Provides high performance, modularity, and scalability


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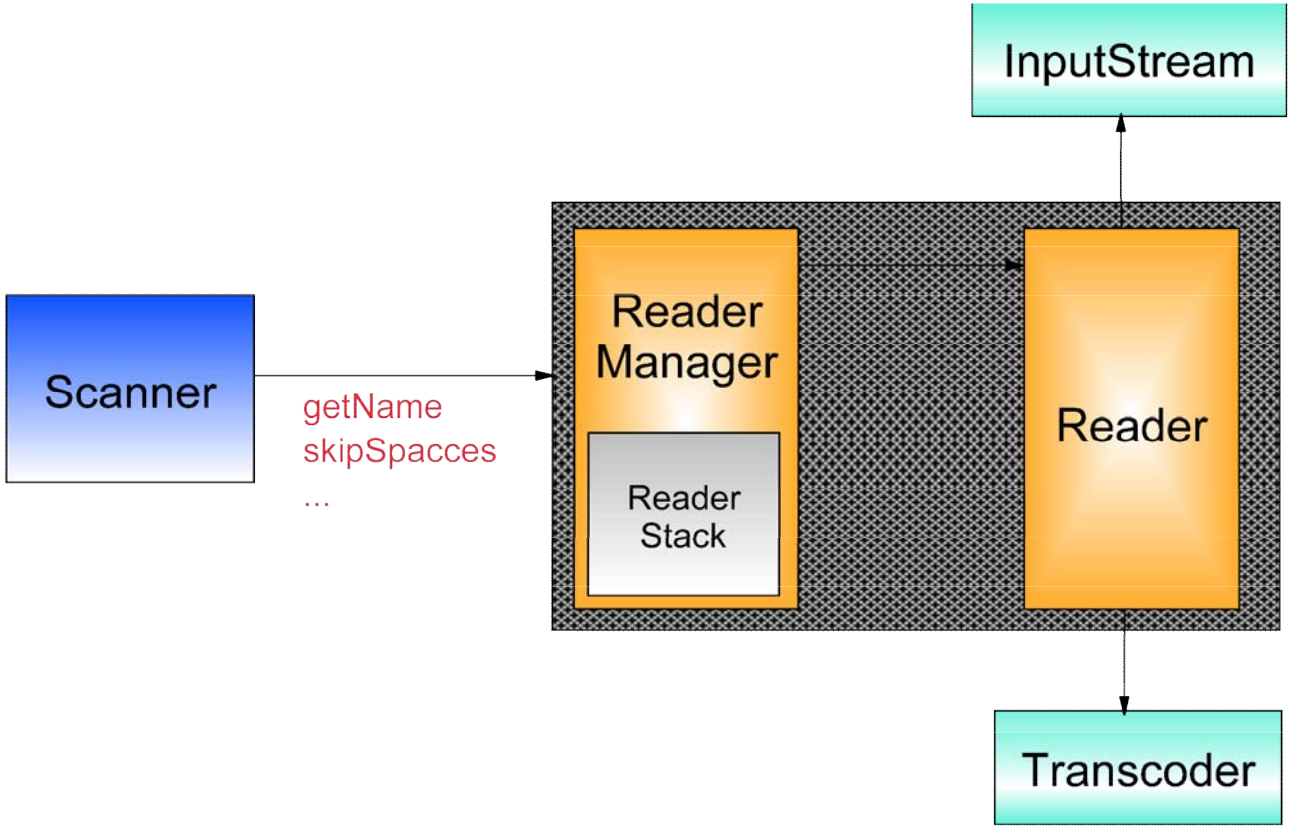
# Parsing Pipeline





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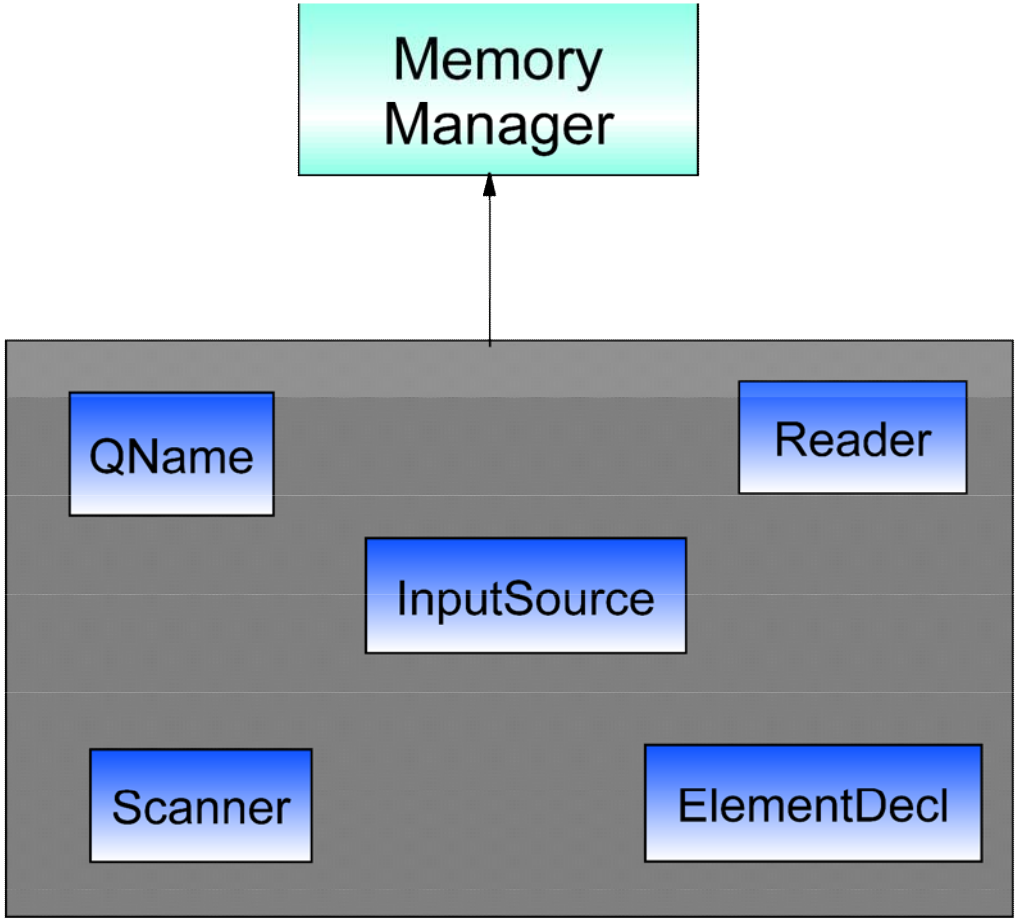
# Reader Management





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# Pluggable Memory Management




Xerces-C components


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# Feature Highlight

- Grammar caching and pre-parsing
- Serialization of XML
- Other features


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# Grammar Caching and Pre-parsing

- Xerces-C provides a mechanism to pre-parse grammars

```
XercesDOMParser parser;
parser.setDoSchema(true);
parser.setDoNamespaces(true);
// Let's pre-parse the schema grammar (.xsd) and cache it.
Grammar* grammar = parser.loadGrammar(xmlFile,
Grammar::SchemaGrammarType, true);
…
// Ask to cache automatically
parser.cacheGrammarFromParse(true);
```


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# Grammar Caching and Pre-parsing

- Xerces-C allows applications to re-use previously parsed grammars when parsing new XML documents

```
// Enable grammar caching
parser.cacheGrammarFromParse(true);

// parser the document
parser.parse(xmlFile);

// Enable re-use of grammars
parser.useCachedGrammarInParse(true);
```


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# Loading and Serializing using DOM

```
DOMImplementation *impl =
DOMImplementationRegistry::getDOMImplementation("LS");

DOMBuilder* myParser = ((DOMImplementationLS*)impl)->
        createDOMBuilder(DOMImplementationLS::MODE_SYNCHRONOUS, 0);

DOMNode* aDOMNode = myParser->parseURI("theXMLFile.xml");

DOMWriter* myWriter = ((DOMImplementationLS*)impl)->createDOMWriter();
if (myWriter->canSetFeature(XMLUni::fgDOMWRTBOM, true))
    myWriter->setFeature(XMLUni::fgDOMWRTBOM, true);

XMLCh* theXMLString_Unicode = myWriter->writeToString(*aDOMNode);

XMLString::release(&theXMLString_Unicode);
myWriter->release();
myParser->release();
```


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# Other Features

- Progressive parsing
- Pluggable transcoders


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# References

- http://xml.apache.org/xerces2-j/
- http://xml.apache.org/xerces-c/index.html
- http://www.alphaworks.ibm.com/tech/xml4j
- http://www.apache.org/~andyc/xml/present/


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